IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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9 October 2026



Executive summary • Business problem and data sources
Methodology • API, scraping, wrangling, SQL, EDA, geoanalytics, Dash, models
Results • Observed data patterns, spatial context and classification
Conclusion • Findings, uncertainty, limitations and recommendations
Appendix • Reproducibility notes and provenance

Objective: examine drivers of Falcon 9 first-stage landing outcomes and demonstrate reproducible analysis.
90-launch analytical cohort (2010–2020); 60 successful landings, 30 other outcomes.
Observed site-specific landing rates range from 60.0% to 77.3%.
Model comparison uses stratified 5-fold validation on a training split; highest CV accuracy: SVM (RBF) (88.9%).
Source: historical IBM-course research datasets; analyses reproduced from documented data extracts.

Commercial launch economics depend materially on recovering and reusing first-stage hardware.
Research question: which observed mission characteristics align with landing success, and how reliably can outcomes be predicted?
Scope: course-style historical samples, not a real-time operational forecast.
Analytic unit: one recorded first-stage launch; success is coded as Class = 1.

Section 1

Methodology



1 DATA INGESTION SpaceX API exercise and historical web-table records
2 WRANGLING Normalize fields, define binary landing class, inspect missingness
3 EXPLORATION Visual comparisons by launch site, orbit, payload, and time
4 SPATIAL / DASHBOARD Location plotting and interactive-style aggregations
5 PREDICTION Stratified training/holdout; cross-validated models
6 AUDIT Dataset and code provenance, caveats and reproducibility

Reference material: published course-style SpaceX Falcon 9 datasets and notebooks.
Independent reconstruction uses the included CSV and SQLite assets; no live API requests are claimed.
Analytic sources: dataset_part_2.csv (90 records), SPACEXTBL (SQL), spacex_launch_geo.csv and spacex_launch_dash.csv (56-record subsets).
Different extracts have different coverage and occasionally inconsistent fields; results are never merged without qualification.

Data Collection — SpaceX API

Analytical data lineage



Collection workflow documents the source data structure and field transformations.

Source: third-party reference archive supplied for independent reconstruction. Dataset descriptions and technical details are reproduced with attribution.

Data Collection - Scraping

Analytical data lineage



Reference workflow based on published capstone methodology.

Source: third-party reference archive supplied for independent reconstruction. Dataset descriptions and technical details are reproduced with attribution.

Data integrity checks: record counts, column typing, category values, and missingness.
Target variable: Class = 1 for recorded successful first-stage landing; 0 otherwise.
90 records; positive class 60 (66.7%), negative class 30 (33.3%).
Missing LandingPad values (26) are not imputed as evidence of landing failure.
For modeling, eight numerical/binary predictors are used; missing numeric values receive zero only where applicable.
Separate SQL and 56-record spatial/dashboard sources retain their own denominator.

Scatter plots: flight number and payload against site/orbit, coloured by landing outcome.
Bar charts: success rates by orbit; category labels include group size.
Line series: annual share of successfully landed boosters from 2010–2020.
Interpretation: temporal and site associations do not establish causality; unequal counts limit comparisons.
Charts on slides 18–23 are independently generated from dataset_part_2.csv.

SQLite inspection confirms SPACEXTBL table and landing/mission outcome fields.
Queries use DISTINCT, LIKE, BETWEEN, MIN, AVG, SUM, GROUP BY, and ORDER BY.
Results cover sites, NASA payload mass, booster version, ground/drone-ship outcomes and rankings.
SQLite database contains historical date formatting quirks; treat ambiguous date results cautiously.
Slides 24–33 present reproducible SQL query results.

Spatial reference: 56 geolocated launch records with latitude, longitude and landing class.
Use coordinate markers to distinguish launches and locations.
Use circles or symbol sizes to show launch frequency by site.
Compare observed success share across sites; sample sizes must be displayed.
Slides 35–37 present static spatial analysis views.

Dashboard objective: filter flights by launch site and payload band to compare landing outcomes.
Site success pie chart; payload-versus-outcome scatter; success-versus-other grouped comparison.
A site selector and payload-mass range control are appropriate interactive elements.
Slides 39–41 show static analytical charts derived from the 56-record dashboard data extract.

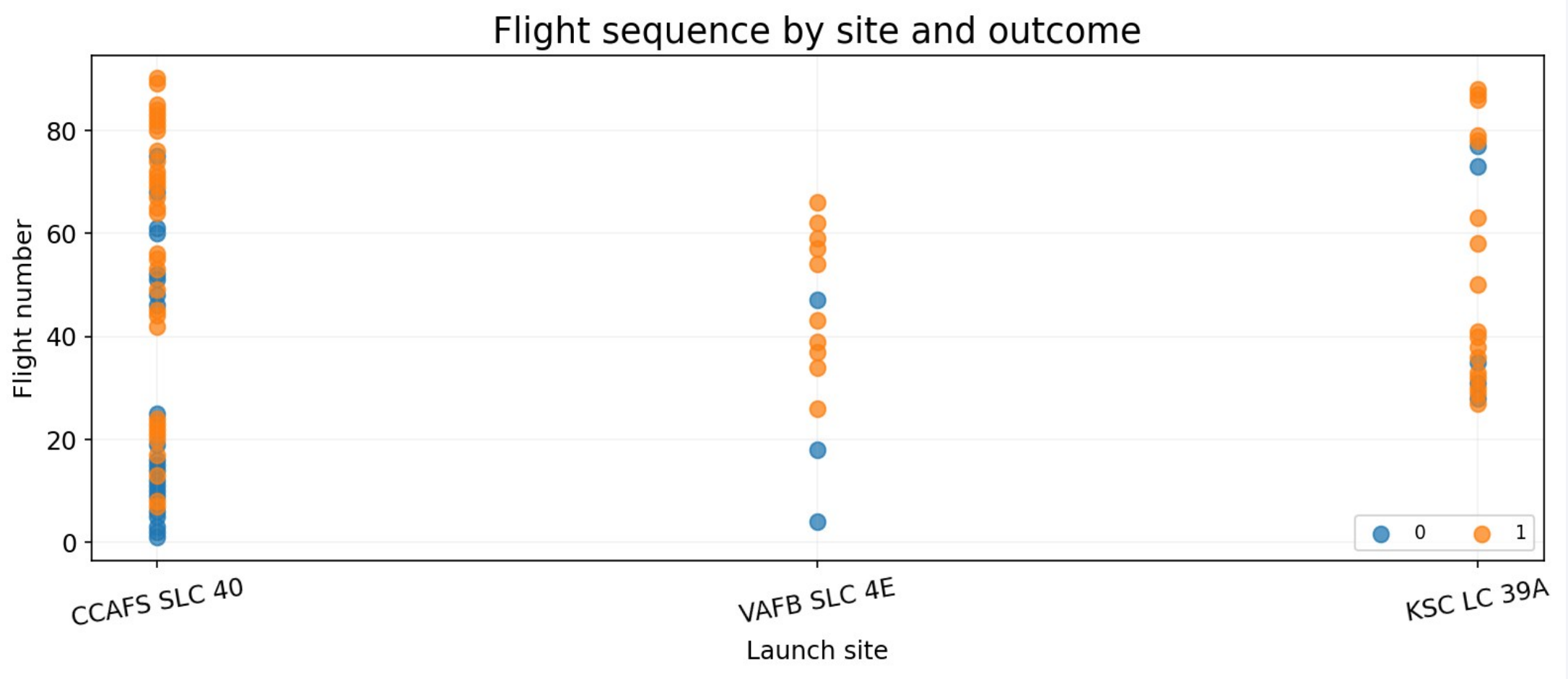
Design: 90-record cohort; 80/20 stratified train/test split (random_state=42).
Predictors: flight order, payload, flights, grid fins, reused, legs, block and reuse count.
Models: regularized logistic regression, RBF SVM, shallow decision tree and k-nearest neighbours.
Model choice: training-only 5-fold stratified cross-validation; holdout reserved for final assessment.
Limitation: small, historical nonrandom sample; reported accuracies are illustrative, not deployable.

EXPLORATORY RESULTS Flight sequence, payload, orbit, launch site and annual trends
SQL RESULTS Ten transparent SQLite queries with reported outputs
SPATIAL ANALYSIS Launch-site coordinates and site-level comparisons
DASHBOARD Three independently reconstructed views
PREDICTIVE ANALYSIS Cross-validated model comparison and untouched holdout confusion matrix

Section 2

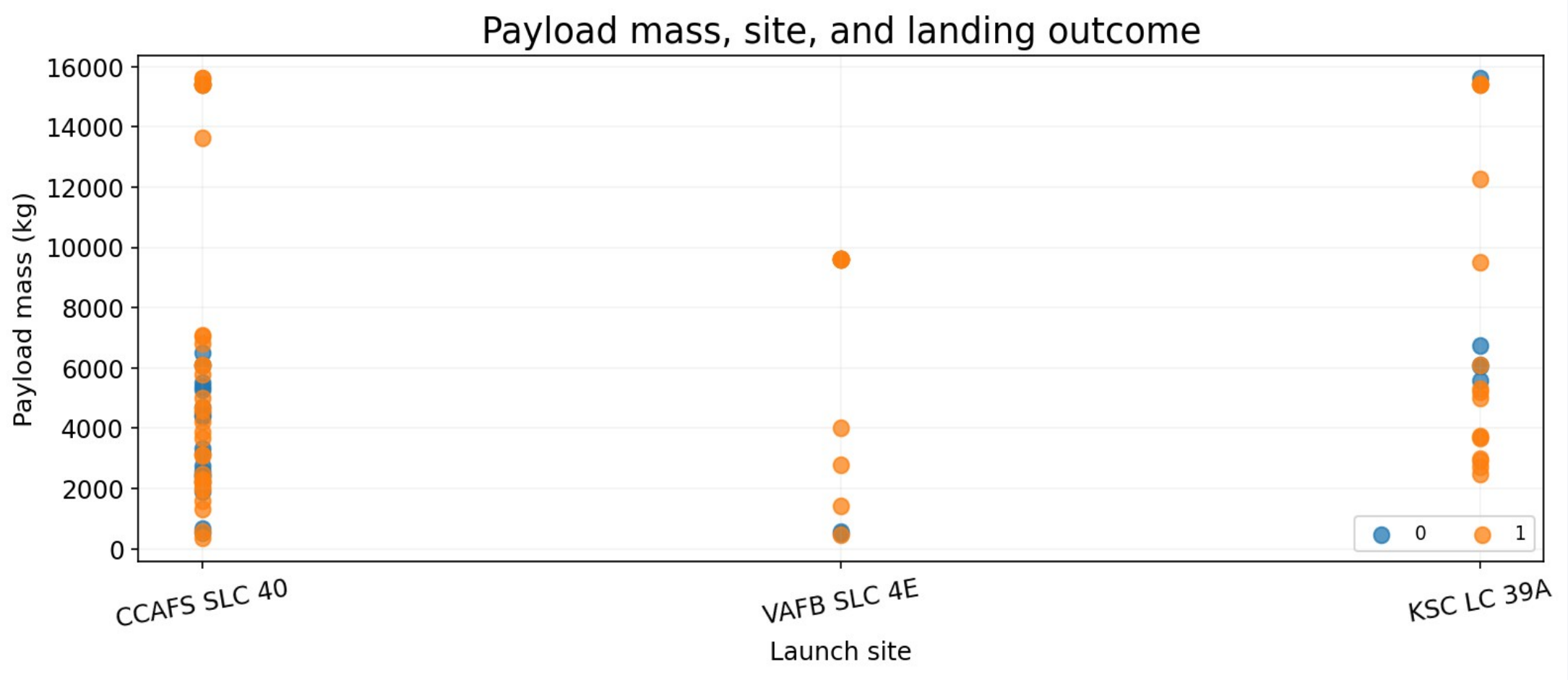
Insights drawn from EDA

Flight Number vs. Launch Site



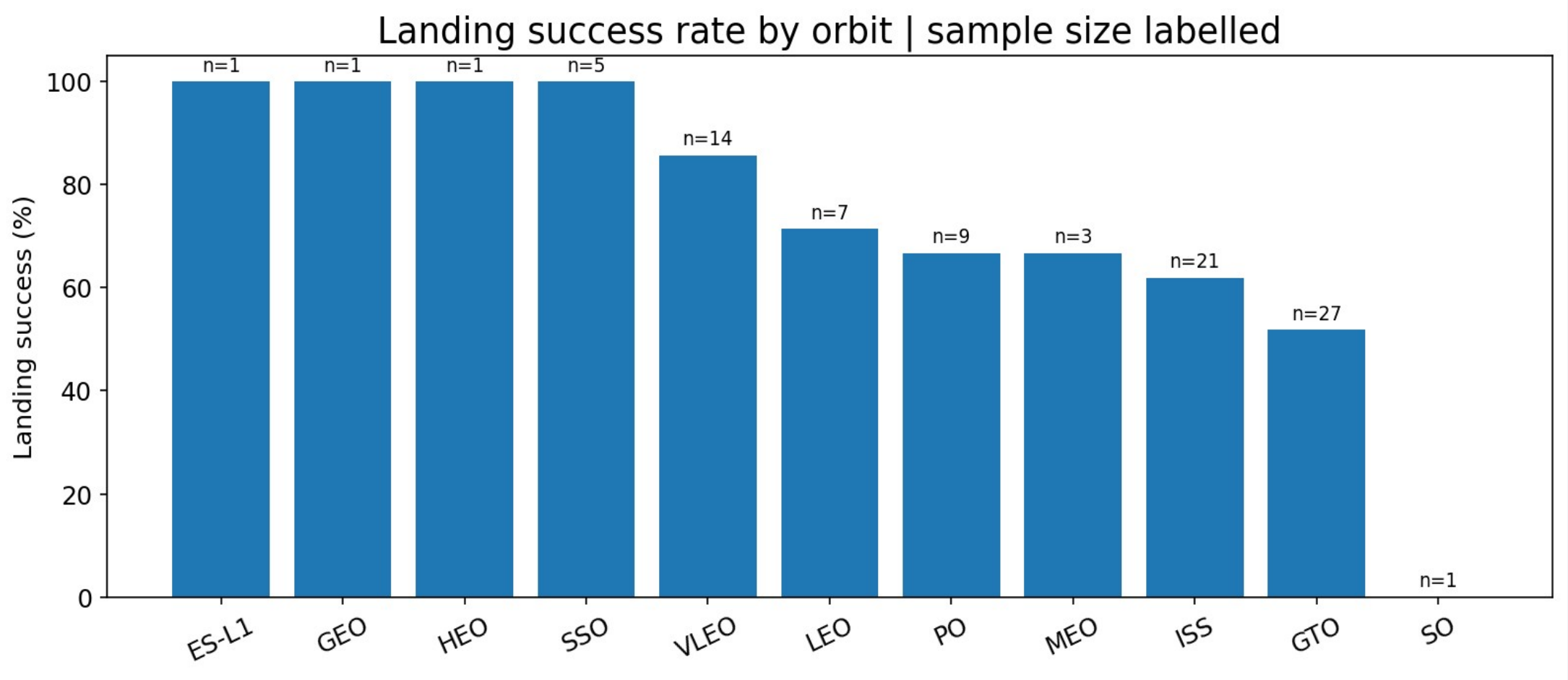
Source: historical SpaceX course dataset (n=90).

Payload vs. Launch Site



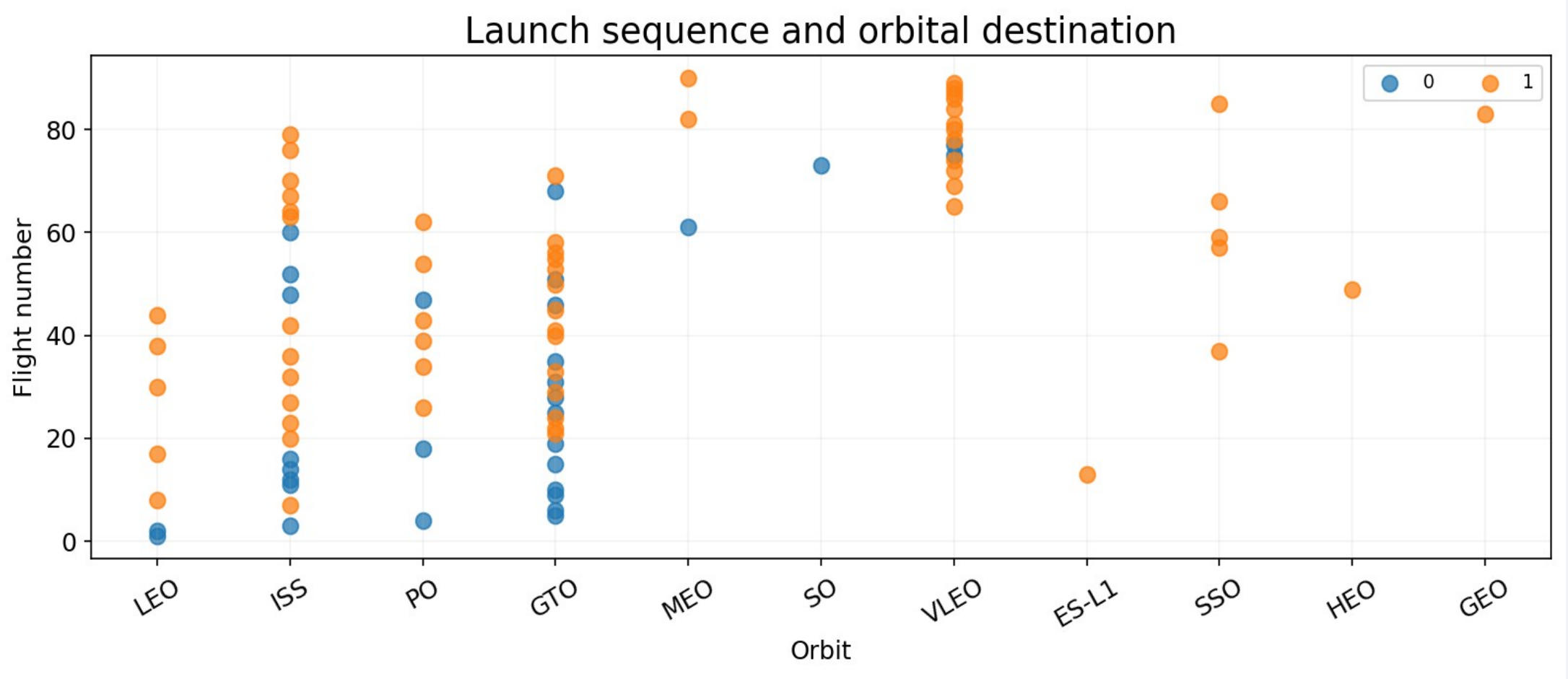
Source: historical SpaceX course dataset (n=90).

Success Rate vs. Orbit Type



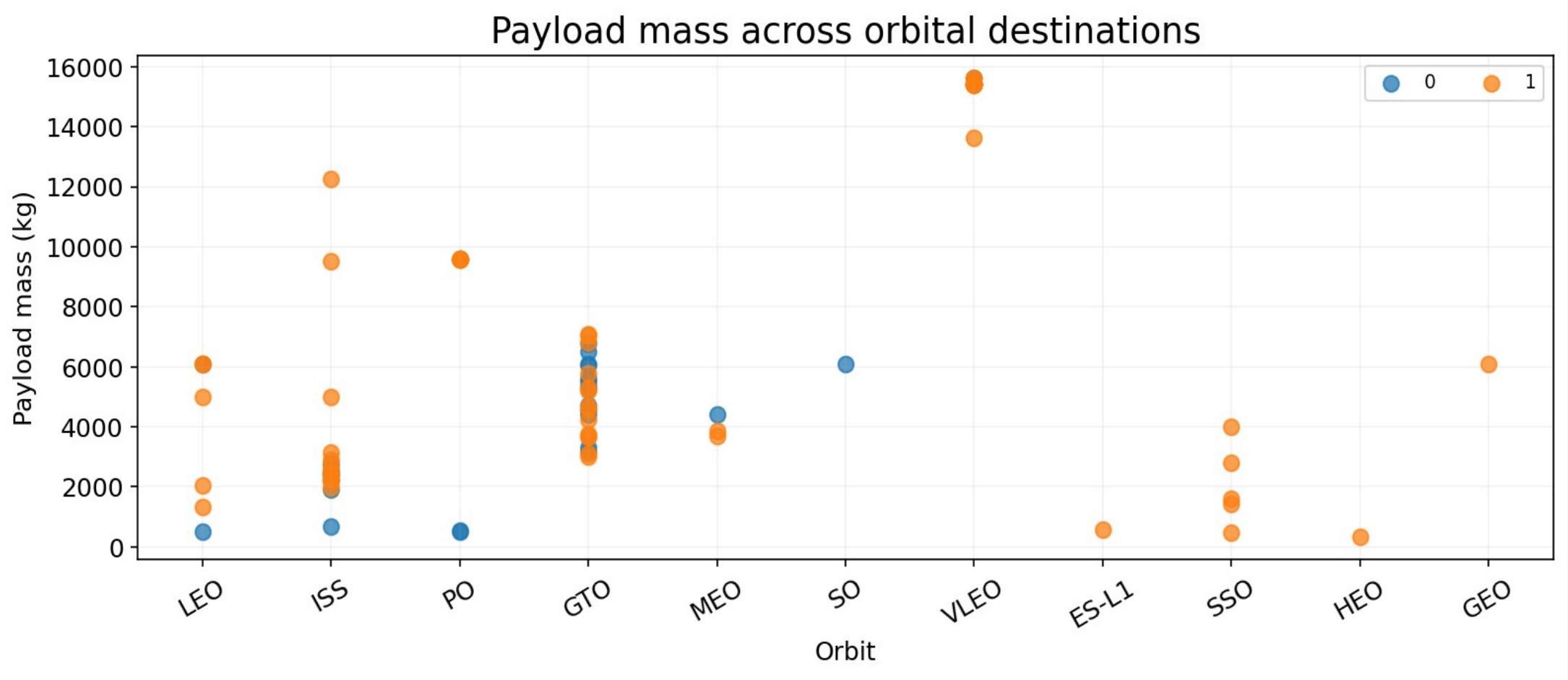
Source: historical SpaceX course dataset (n=90).

Flight Number vs. Orbit Type



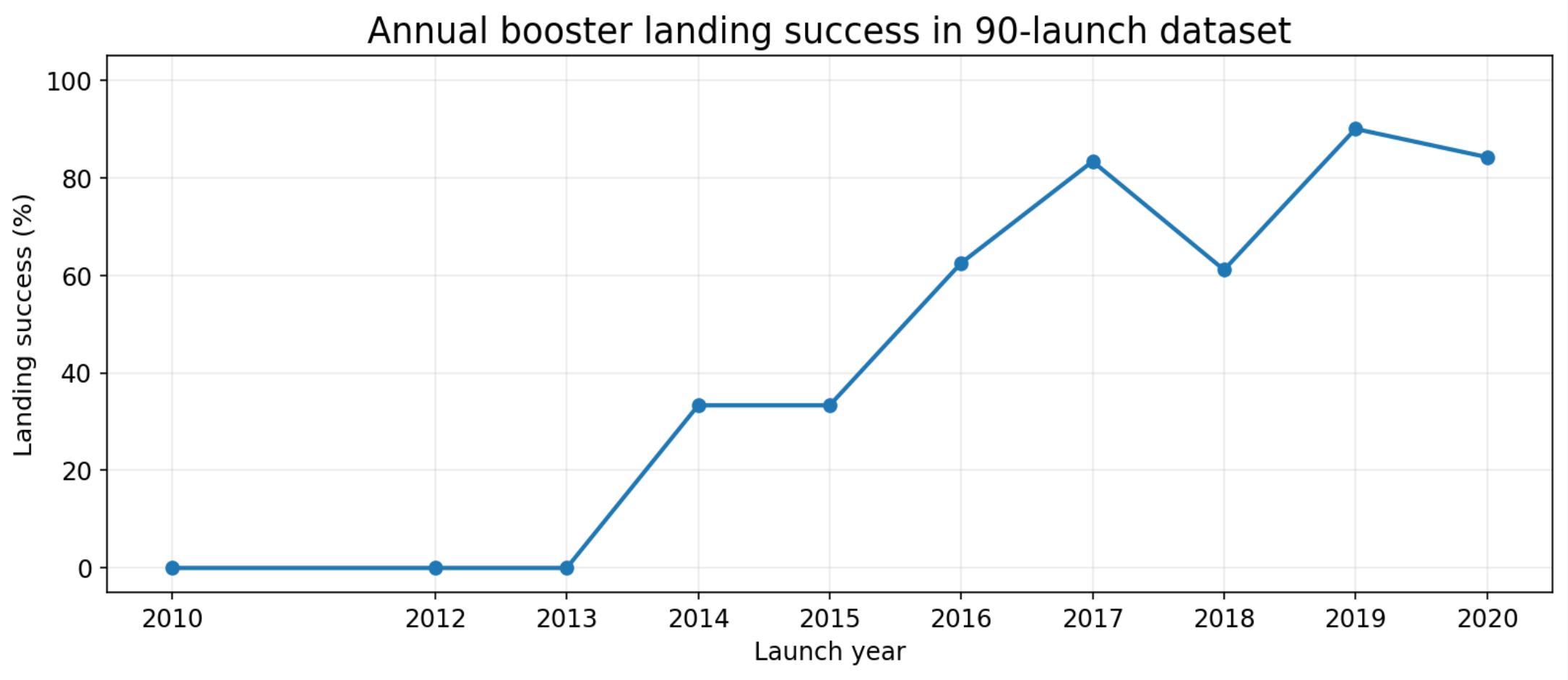
Source: historical SpaceX course dataset (n=90).

Payload vs. Orbit Type



Source: historical SpaceX course dataset (n=90).

Launch Success Yearly Trend



Source: historical SpaceX course dataset (n=90).

All Launch Site Names

SQL query:
SELECT DISTINCT Launch_Site FROM SPACEXTBL

RESULTS (4 returned row(s)):
 Launch_Site
 CCAFS LC-40
 VAFB SLC-4E
  KSC LC-39A
CCAFS SLC-40

Source: historical SpaceX SQLite dataset (SPACEXTBL).

Launch Site Names Begin with 'CCA'

SQL query:
SELECT Launch_Site, Date FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5

RESULTS (5 returned row(s)):
Launch_Site Date
CCAFS LC-40 2010-04-06
CCAFS LC-40 2010-08-12
CCAFS LC-40 2012-05-22
CCAFS LC-40 2012-08-10
CCAFS LC-40 2013-01-03

Source: historical SpaceX SQLite dataset (SPACEXTBL).

Total Payload Mass

SQL query:
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_kg FROM SPACEXTBL WHERE Customer LIKE '%NASA%'

RESULTS (1 returned row(s)):
 Total_kg
  107010

Source: historical SpaceX SQLite dataset (SPACEXTBL).

Average Payload Mass by F9 v1.1

SQL query:
SELECT AVG(PAYLOAD_MASS__KG_) AS Mean_kg FROM SPACEXTBL WHERE Booster_Version LIKE '%F9 v1.1%'

RESULTS (1 returned row(s)):
   Mean_kg
2534.666667

Source: historical SpaceX SQLite dataset (SPACEXTBL).

First Successful Ground Landing Date

SQL query:
SELECT MIN(Date) AS First_Date FROM SPACEXTBL WHERE Landing_Outcome LIKE '%Success%ground%'

RESULTS (1 returned row(s)):
First_Date
2015-12-22

Source: historical SpaceX SQLite dataset (SPACEXTBL).

Successful Drone Ship Landing with Payload between 4000 and 6000

SQL query:
SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome LIKE '%Success%drone%' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000

RESULTS (4 returned row(s)):
Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Source: historical SpaceX SQLite dataset (SPACEXTBL).

Total Number of Successful and Failure Mission Outcomes

SQL query:
SELECT Mission_Outcome, COUNT(*) AS N FROM SPACEXTBL GROUP BY Mission_Outcome

RESULTS (4 returned row(s)):

Mission_Outcome	N
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Source: historical SpaceX SQLite dataset (SPACEXTBL).

Boosters Carried Maximum Payload

SQL query:
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL ORDER BY PAYLOAD_MASS__KG_ DESC LIMIT 5

RESULTS (5 returned row(s)):

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600

Source: historical SpaceX SQLite dataset (SPACEXTBL).

2015 Launch Records

SQL query:
SELECT Date, Booster_Version, Landing_Outcome FROM SPACEXTBL WHERE Landing_Outcome LIKE '%Failure%drone%' AND SUBSTR(Date,1,4)='2015'

RESULTS (2 returned row(s)):

Date	Booster_Version	Landing_Outcome
2015-10-01	F9 v1.1 B1012	Failure (drone ship)
2015-04-14	F9 v1.1 B1015	Failure (drone ship)

Source: historical SpaceX SQLite dataset (SPACEXTBL).

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

SQL query:
SELECT Landing_Outcome, COUNT(*) AS N FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY N DESC

RESULTS (8 returned row(s)):

Landing_Outcome	N
No attempt	10
Success (ground pad)	5
Success (drone ship)	5
Failure (drone ship)	5
Controlled (ocean)	3
Uncontrolled (ocean)	2
Precluded (drone ship)	1
Failure (parachute)	1

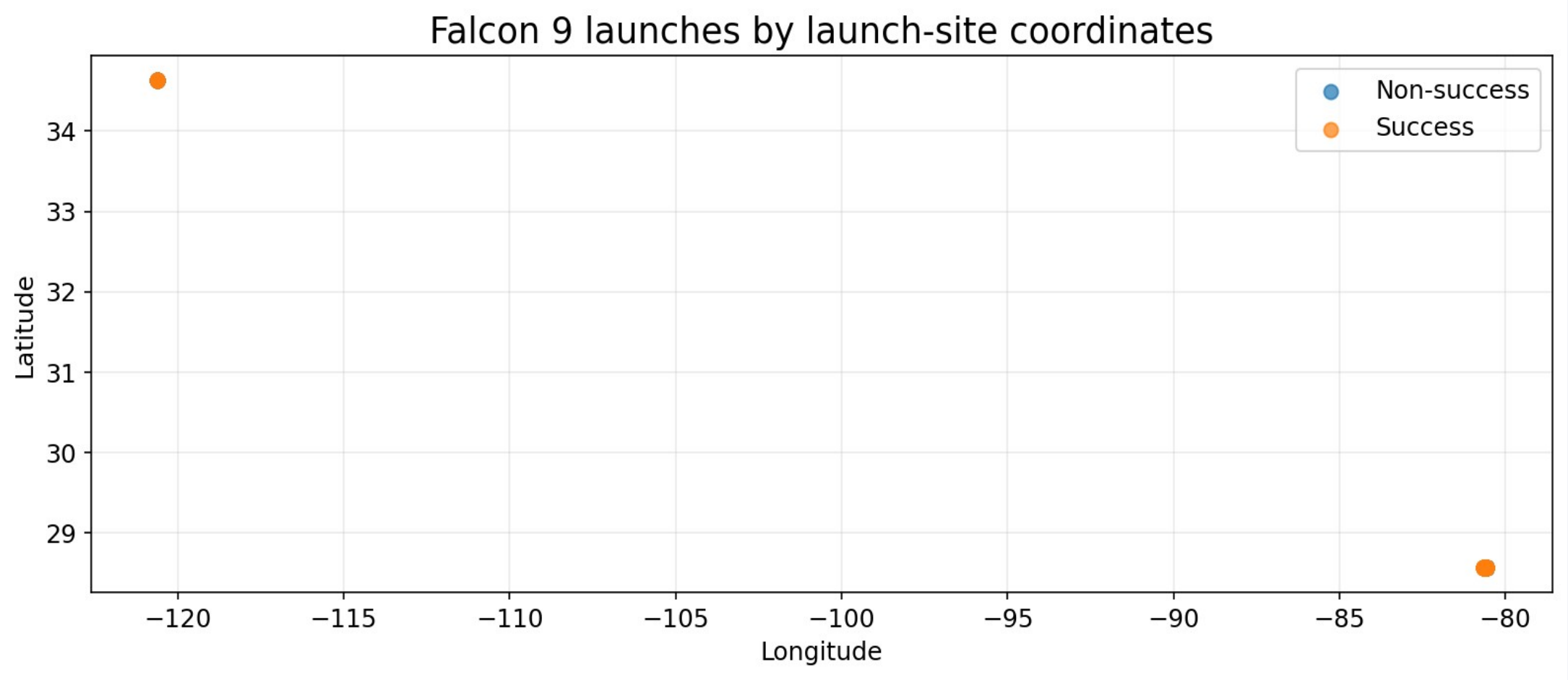
Source: historical SpaceX SQLite dataset (SPACEXTBL).



Section 3

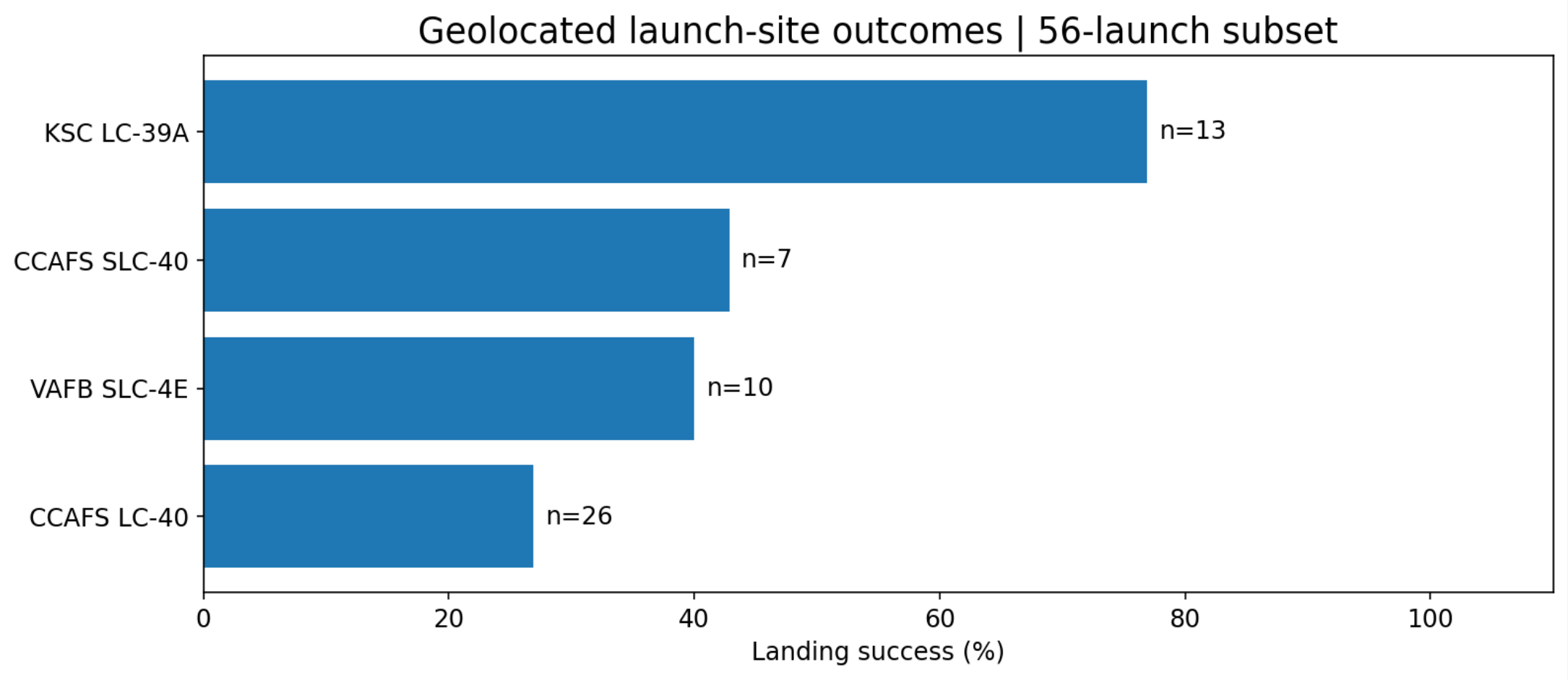
Launch Sites Proximities Analysis

<Folium Map Screenshot 1>



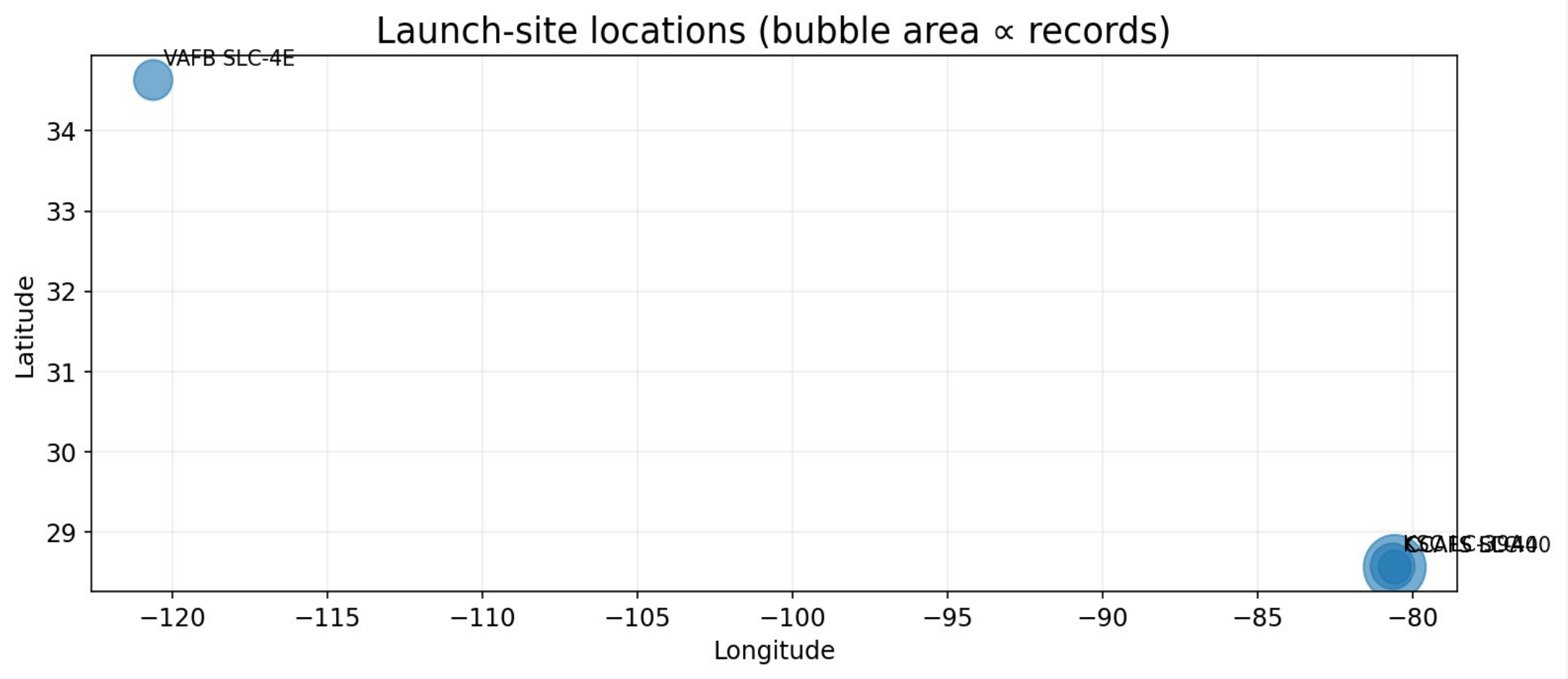
Source: historical SpaceX launch-site dataset (n=56).

<Folium Map Screenshot 2>

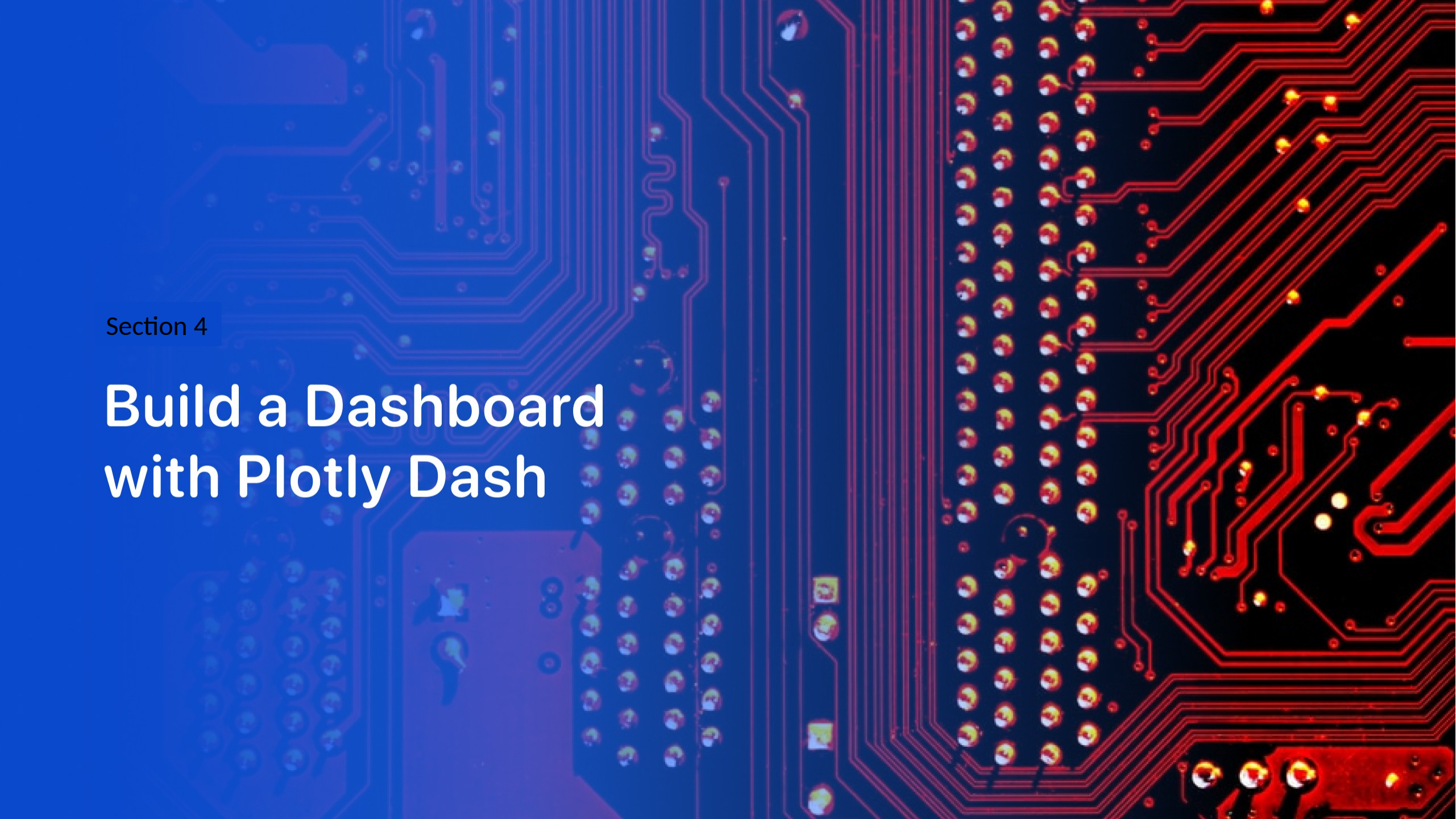


Source: historical SpaceX launch-site dataset (n=56).

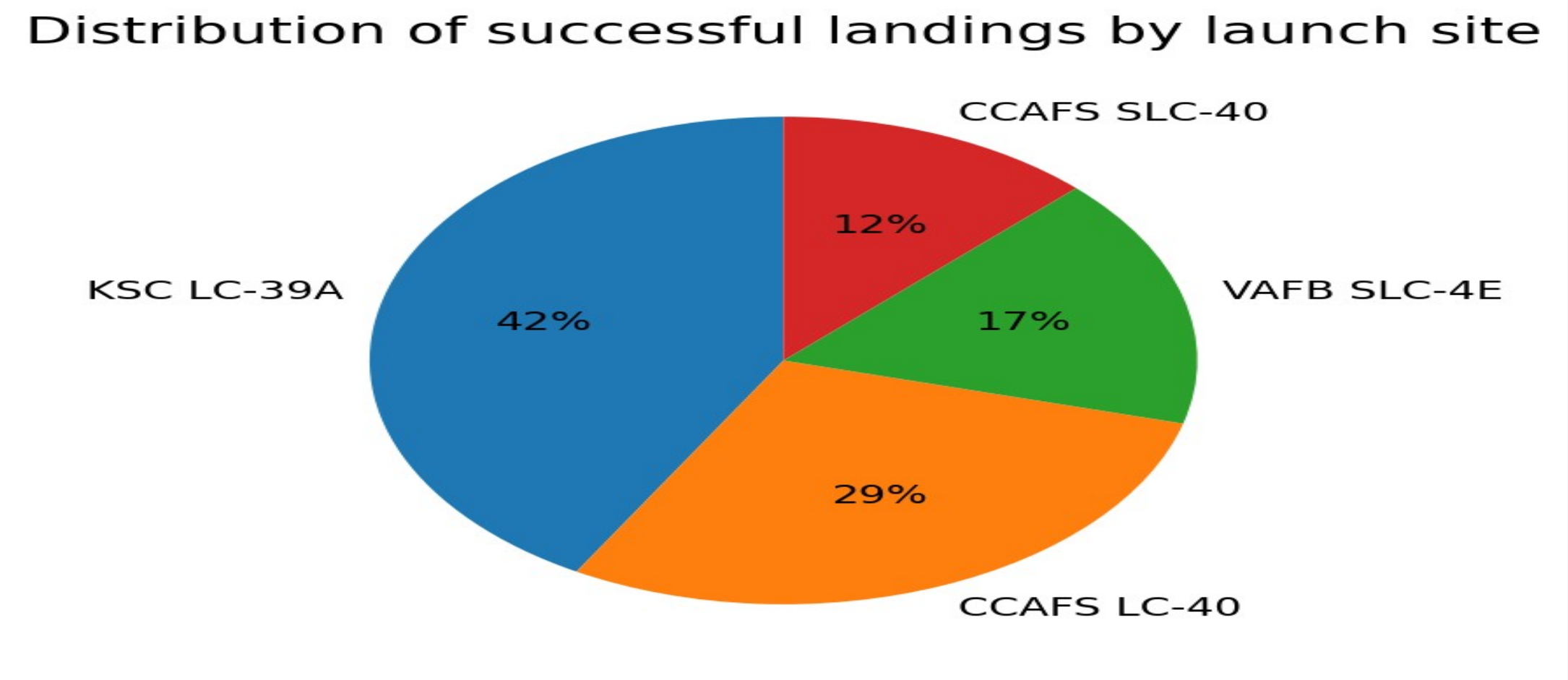
<Folium Map Screenshot 3>



Source: historical SpaceX launch-site dataset (n=56).

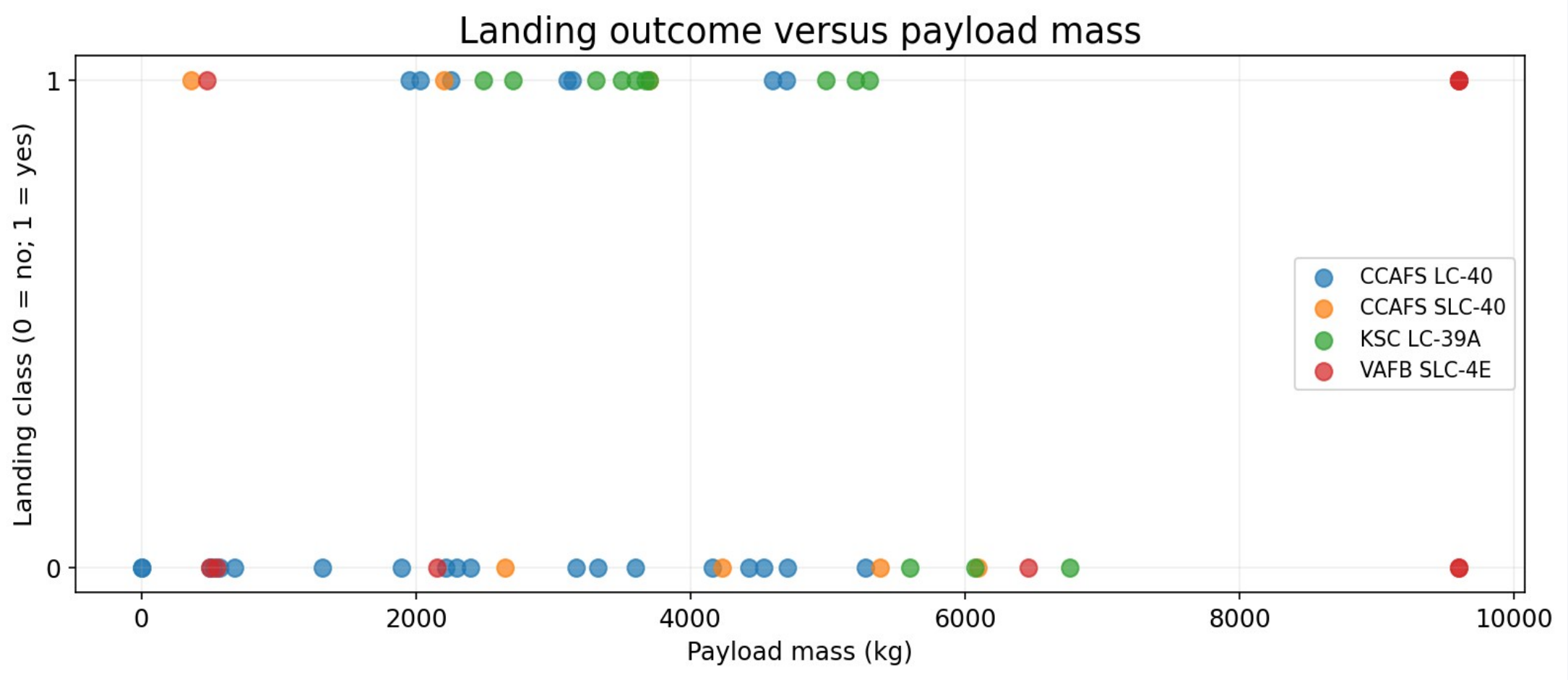

Section 4
Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>



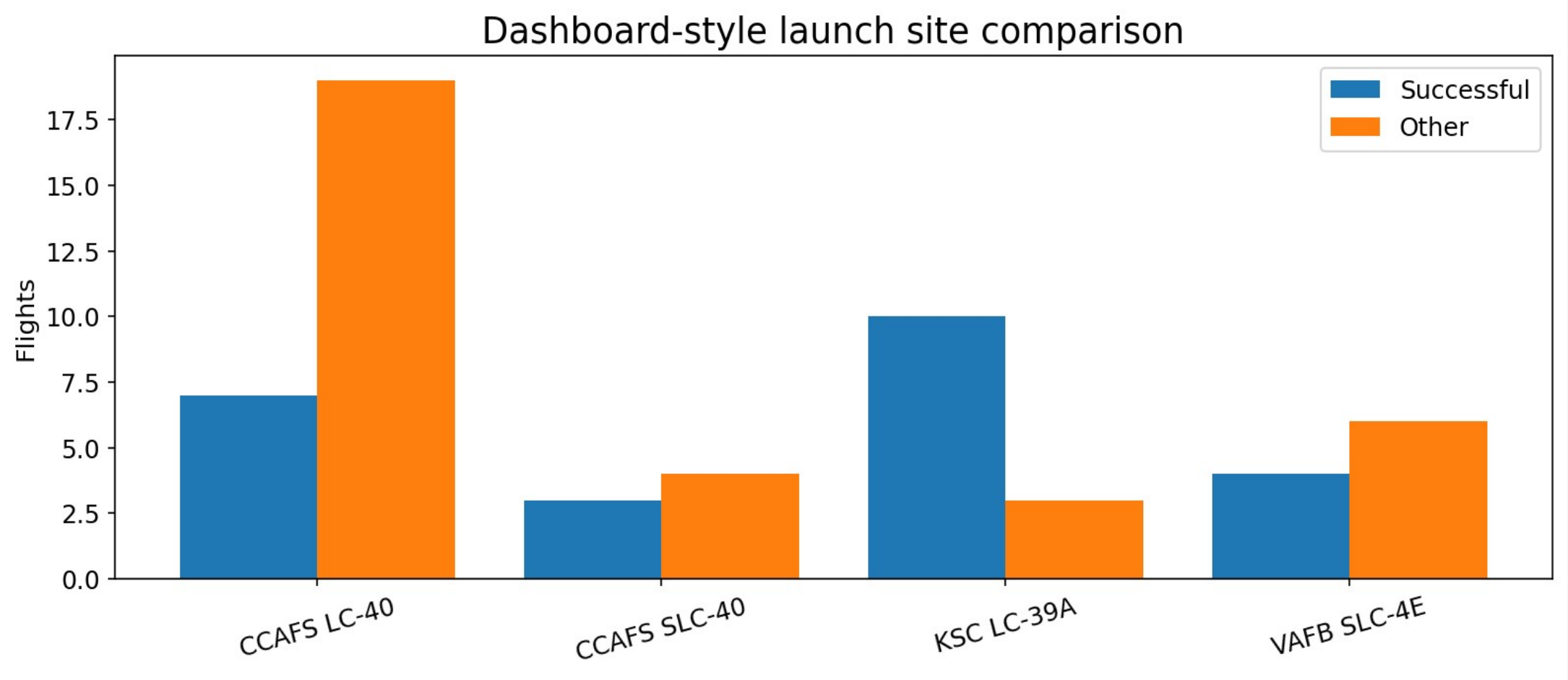
Source: historical SpaceX launch-site dataset (n=56).

<Dashboard Screenshot 2>



Source: historical SpaceX launch-site dataset (n=56).

<Dashboard Screenshot 3>



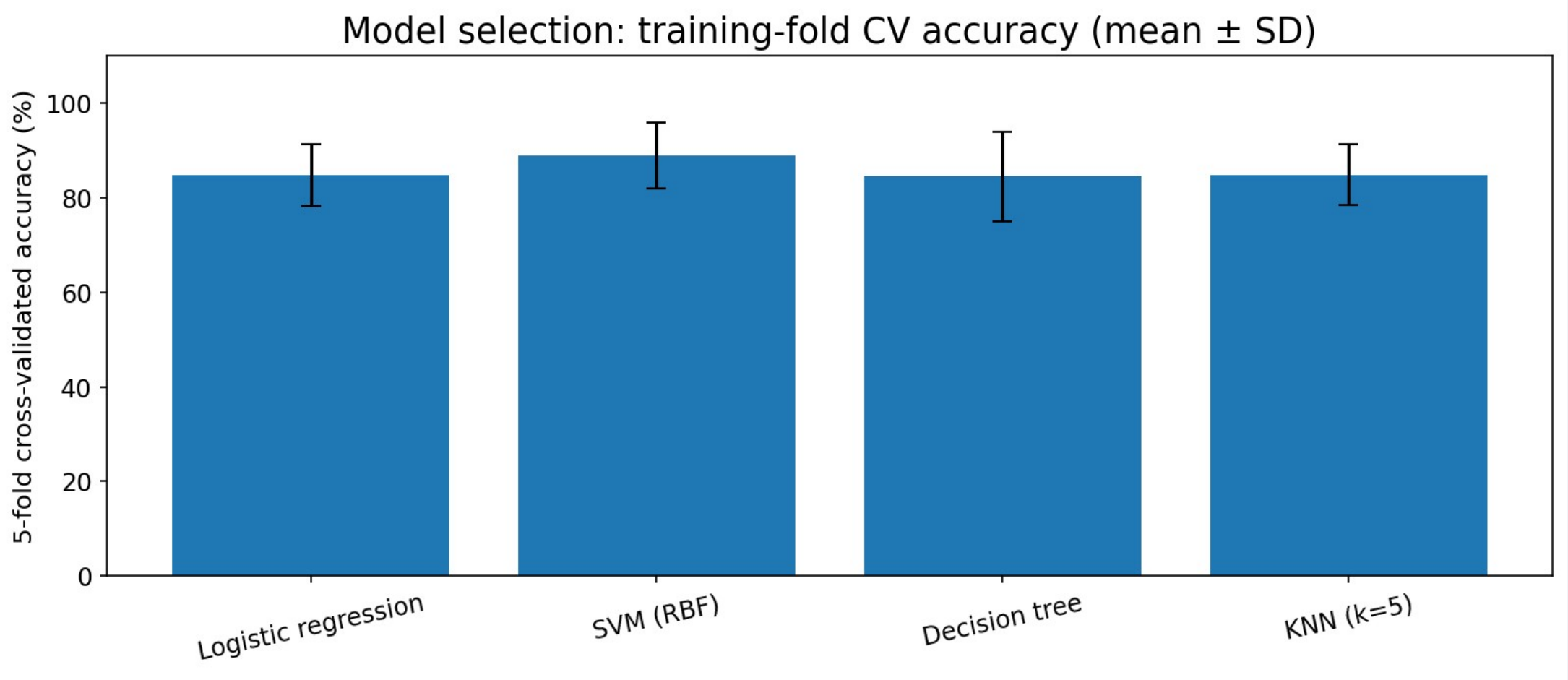
Source: historical SpaceX launch-site dataset (n=56).

Section 5

Predictive Analysis (Classification)

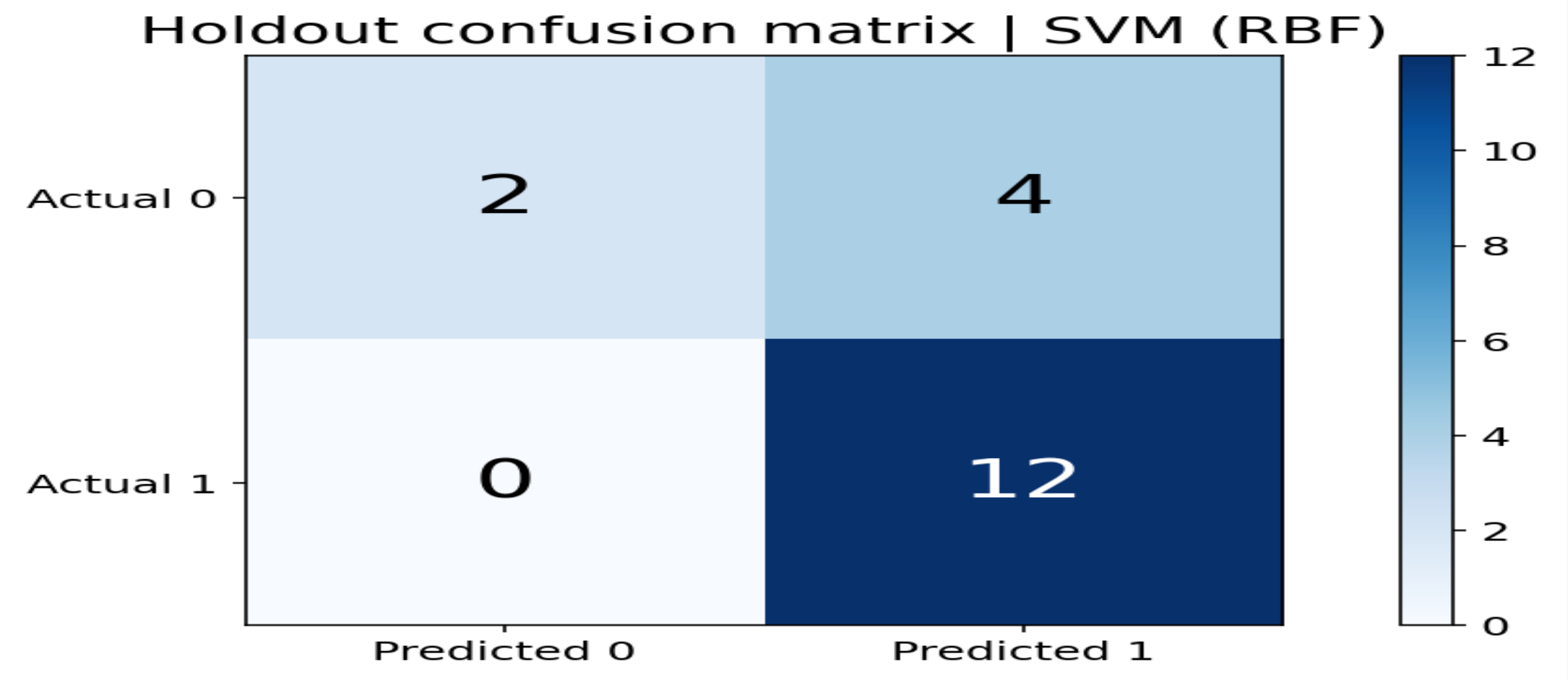


Classification Accuracy



Source: historical SpaceX course dataset (n=90).

Confusion Matrix



Source: historical SpaceX course dataset (n=90).

1 Landing success in the 90-launch sample is 66.7% (60/90); baseline accuracy is therefore already 66.7%.
2 Launch site, orbit, payload and time are associated with outcomes, but small subgroup samples constrain inference.
3 Best training-fold model: SVM (RBF) (88.9% CV); holdout accuracy 77.8% (18 cases).
4 Model reliability requires larger temporally held-out datasets, calibration and external validation.
5 Dataset coverage, sampling constraints and source provenance should inform all operational interpretations.

DATA SOURCES

Third-party IBM course-style reference archive: datasets, Jupyter notebooks and SQLite database supplied for this reconstruction.

Course grading template: IBM Skills Network, “Winning Space Race with Data Science,” 47-slide template.

REPRODUCIBILITY

Analytic cohort: dataset_part_2.csv; static dashboard source: spacex_launch_dash.csv; locations: spacex_launch_geo.csv; SQL: my_data1.db.

Repository created for this project: https://github.com/marvindebo/IBM-Applied-Data-Science-Capstone.

Data limitations and reproducibility

Data sources: historical SpaceX capstone datasets and the associated SQLite query archive.
The 90-record classification dataset and 56-record geospatial extract cover different sample frames.
Historical SQLite records contain date-format inconsistencies; date-based comparisons require caution.
Small site-level samples limit statistical inference and generalizability.
Further work: temporal validation, expanded launch history, calibration and sensitivity analysis.

Thank you!

